

Relational Kernels: Geometry, Bell Nonlocality, and Formal Bounds for the Geometry-CHSH Tradeoff

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Updated theorem draft - incorporating exact noisy-Tsirelson completion and formal $C^*(\mu)$ status
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Abstract

We study normalized positive semidefinite (PSD) relational kernels as common-origin structures from which geometric and Bell-correlation properties may be derived. A symmetric normalized PSD kernel G can be read as a Gram matrix of unit vectors, while the transformation $d(i,j) = -\log |G(i,j)|$ produces a distance-like object whenever the kernel obeys a multiplicative consistency condition. This condition, called μ -submultiplicativity, says that relational strength must compose across triples: $|G(i,k)| \geq \mu |G(i,j)| |G(j,k)|$.

The paper establishes a qualitative obstruction: exact Tsirelson saturation in the CHSH scenario is incompatible with any nonzero uniform μ -submultiplicativity on the Bell sector. We further introduce an observable completion invariant $\mu_{\max}(E)$, defined from a CHSH correlator table E by maximizing geometric consistency over all PSD Gram completions. Along the canonical noisy-Tsirelson family $E(x) = [[x, x], [x, -x]]$, the completion problem is solved exactly: $\mu_{\max}(E(x)) = \min(1, (\sqrt{1-x^2} - x)/x^2)$, with the endpoint interpreted by continuity at $x=0$.

New machine-checked results also formalize the global $C^*(\mu)$ problem: feasible sets shrink monotonically with μ , $C^*(\mu)$ is decreasing, extremizers exist, and the strict obstruction $C^*(\mu) < 2\sqrt{2}$ holds for every $\mu > 0$. A conditional quantitative gap $S^2 \leq 8 - \mu^2/8$ is proved in the large-correlator regime. The same formalization disproves a tempting symmetric ansatz: the matrix giving the often-suggested value $4/\sqrt{2+\mu}$ is not PSD for $\mu > 0$. Thus the full $C^*(\mu)$ curve remains open and requires asymmetric or higher-rank constructions.

1. Introduction

Bell nonlocality shows that quantum correlations can violate classical local-hidden-variable constraints. In the CHSH scenario, classical correlations satisfy $|S| \leq 2$, while quantum correlations satisfy the Tsirelson bound $|S| \leq 2\sqrt{2}$. This paper asks how such nonlocal correlations interact with a different structural requirement: the requirement that relational strengths compose consistently enough to support emergent geometry.

The organizing object is a normalized PSD kernel G . Such a kernel is equivalent to a Gram matrix of unit vectors. The same object can therefore be interpreted in two ways: as a quantum-adjacent table of inner products and as a candidate source of distance by setting $d(i,j) = -\log |G(i,j)|$. The bridge from relation to geometry is μ -submultiplicativity: $|G(i,k)| \geq \mu |G(i,j)| |G(j,k)|$. This condition is exactly the multiplicative form of an approximate triangle inequality.

The main conceptual point is not that geometry and Bell nonlocality are mutually exclusive. They are not. Rather, they form a spectrum. Weak and moderate Bell-type correlations may be compatible with positive geometric consistency, while maximal CHSH nonlocality forces the compatible geometric parameter to vanish.

This draft incorporates two substantial robustness upgrades. First, the observable invariant $\mu_{\max}(E)$ makes the construction applicable to Bell correlator tables without assuming unobserved same-party overlaps. Second, the formal $C^*(\mu)$ analysis clarifies what is proved, what is quantitatively bounded, and what remains open.

2. Relational Kernel Framework

Definition 2.1 (Normalized PSD kernel). A kernel G on a finite index set I is normalized PSD if G is real symmetric, positive semidefinite, and $G(i,i)=1$ for all i in I .

Definition 2.2 (Geometric consistency). For μ in $[0,1]$, G is μ -submultiplicative on a subset T of indices if

$$|G(i,k)| \geq \mu |G(i,j)| |G(j,k)| \quad \text{for all } i,j,k \text{ in } T.$$

Definition 2.3 (Intrinsic consistency parameter). For a finite kernel G , define $\mu(G)$ as the minimum of $|G(i,k)|/(|G(i,j)| |G(j,k)|)$ over triples with nonzero denominator, capped at 1 when used as a consistency parameter.

The log-distance $d(i,j)=-\log |G(i,j)|$ satisfies

$$d(i,k) \leq d(i,j) + d(j,k) - \log(\mu)$$

whenever G is μ -submultiplicative. Thus $\delta=-\log(\mu)$ is a triangle-inequality defect. Exact 1-submultiplicativity gives an exact triangle inequality, subject to the usual caveats about zeros and point separation.

3. Bell Sector and General Obstruction

In the CHSH sector $T=\{A_0,A_1,B_0,B_1\}$, define $x_{ab}=G(A_a,B_b)$ and $S=x_{00}+x_{01}+x_{10}-x_{11}$. The standard PSD/Gram formulation gives $|S| \leq 2\sqrt{2}$.

Theorem 3.1 (Geometric obstruction). If a normalized PSD Bell-sector kernel satisfies uniform μ -submultiplicativity with any $\mu>0$, then it cannot attain $|S|=2\sqrt{2}$.

Proof sketch. Tsirelson saturation forces an orthogonality relation such as $G(A_0,A_1)=0$. Submultiplicativity then gives $|G(A_0,A_1)| \geq \mu |G(A_0,B_b)| |G(B_b,A_1)|$. Since the left side is zero and $\mu>0$, at least one cross-correlator in each relevant product must vanish, which is incompatible with the Tsirelson pattern. The full argument is machine-checked in Lean.

This theorem is qualitative. It proves that maximal Bell violation and nonzero uniform geometric transitivity cannot coexist on the same four-point Bell sector. It does not determine the full quantitative tradeoff curve $C^*(\mu)$.

4. PSD-Completion Diagnostic for Observed Bell Data

Real CHSH experiments provide the four cross-correlators $E_{00}, E_{01}, E_{10}, E_{11}$. They do not directly provide same-party Gram entries such as $G(A_0, A_1)$ and $G(B_0, B_1)$. This motivates an observable completion invariant.

$$G(E, y, z) = [[1, z, E_{00}, E_{01}], [z, 1, E_{10}, E_{11}], [E_{00}, E_{10}, 1, y], [E_{01}, E_{11}, y, 1]].$$

Here $z = G(A_0, A_1)$ and $y = G(B_0, B_1)$ are free completion variables. We require $G(E, y, z)$ to be PSD and define

$$\mu_{\max}(E) = \max_{\{y, z: G(E, y, z) \text{ PSD}\}} \mu(G(E, y, z)).$$

The quantity $\mu_{\max}(E)$ is conservative: it asks whether there exists any geometrically coherent PSD completion compatible with the observed correlators. If even the best completion has small μ , then the Bell data themselves obstruct geometric consistency.

Proposition 4.1 (Observable Tsirelson obstruction). For the Tsirelson-optimal table $E_{00} = E_{01} = E_{10} = 1/\sqrt{2}$, $E_{11} = -1/\sqrt{2}$, one has $\mu_{\max}(E) = 0$.

For real experiments with uncertainty, one can define a robust version $\mu_{\max}^{\epsilon}(E)$ by allowing $|G(A_a, B_b) - E_{ab}| \leq \epsilon_{ab}$. This is the natural form for applying the diagnostic to public Bell-test data.

5. Exact Noisy-Tsirelson Completion Curve

Consider the canonical noisy-Tsirelson family

$$E_{00} = E_{01} = E_{10} = x, \quad E_{11} = -x, \quad 0 \leq x \leq 1/\sqrt{2}.$$

Equivalently, $x = v/\sqrt{2}$, where v in $[0, 1]$ is the usual white-noise visibility and $S = 4x = 2\sqrt{2}v$. The Bell-sector PSD completion is

$$G(y, z) = [[1, z, x, x], [z, 1, x, -x], [x, x, 1, y], [x, -x, y, 1]].$$

Theorem 5.1 (Exact noisy-Tsirelson completion curve). For the above family,

$$\mu_{\max}(E(x)) = \min(1, (\sqrt{1-x^2}-x)/x^2),$$

with the $x=0$ endpoint interpreted by continuity/capping as $\mu_{\max}(E(0))=1$.

5.1 Proof sketch

The determinant of the completed matrix is

$$\det G = (1-y^2)(1-z^2) - 4x^2(1-x^2).$$

PSD implies $\det G \geq 0$. Let $m = \min(|y|, |z|)$. Then $1-y^2 \leq 1-m^2$ and $1-z^2 \leq 1-m^2$, so the determinant inequality gives $(1-m^2)^2 \geq 4x^2(1-x^2)$. For $0 \leq x \leq 1/\sqrt{2}$, this implies

$$m \leq \sqrt{1-x^2} - x.$$

The upper bound is achieved by setting $y = z = t = \sqrt{1-x^2} - x$. The theorem engine proves PSD for this completion via a sum-of-squares decomposition of the quadratic form. Since μ -submultiplicativity requires $\mu x^2 \leq |y|$ and $\mu x^2 \leq |z|$, the formula follows; achievability is verified by finite case analysis over all ordered triples.

5.2 Machine-checked theorem artifacts

- `det_fin_four` - expands the determinant of any 4x4 real matrix into its 24-term Leibniz formula.
- `noisyTsirelsonGram_det` - proves $\det(G) = (1-y^2)(1-z^2) - 4x^2(1-x^2)$.
- `psd_det_upper_bound` - derives the PSD determinant inequality.
- `min_overlap_upper_bound` - proves $\min(|y|, |z|) \leq \sqrt{1-x^2} - x$.
- `noisyTsirelsonGram_achievable_psd` - proves PSD for $y=z=\sqrt{1-x^2}-x$.
- `mu_constraints` - proves that μ -submultiplicativity implies $\mu x^2 \leq |y|$ and $\mu x^2 \leq |z|$.
- `mu_upper_bound` - derives $\mu \leq (\sqrt{1-x^2}-x)/x^2$.
- `mu_achievable` - proves achievability of $\mu = \min(1, t/x^2)$ and full submultiplicativity by finite case analysis.

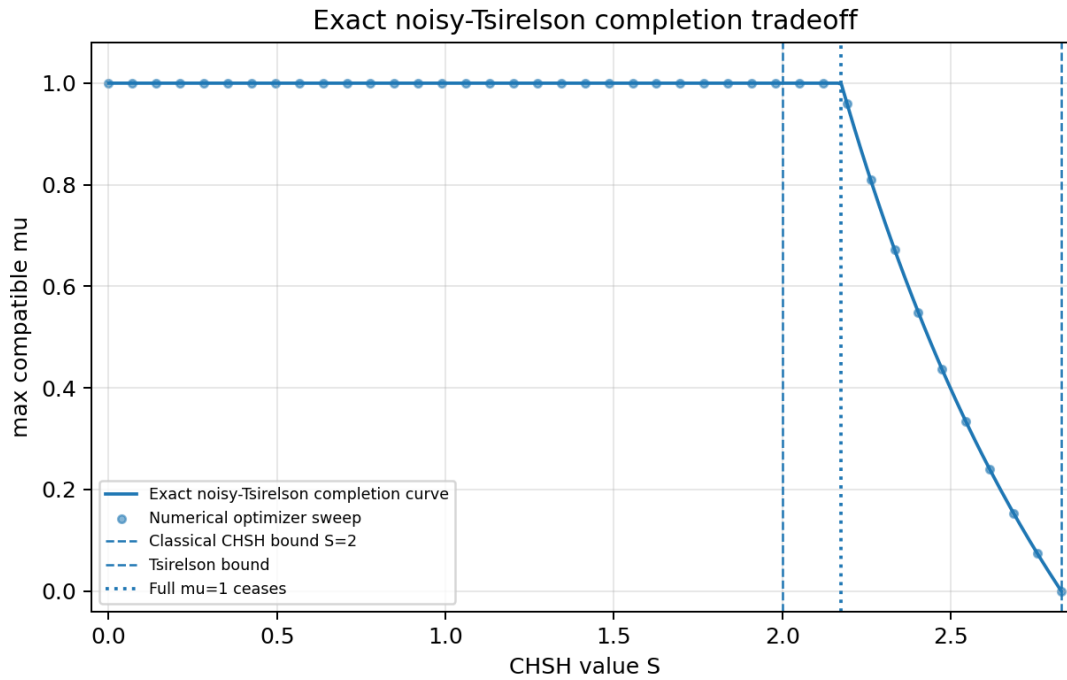


Figure 1. Exact noisy-Tsirelson completion tradeoff. The curve is the closed-form $\mu_{\max}$ formula; numerical optimizer points from the earlier sweep lie on the same curve.

6. Formal Status of the Global $C^*(\mu)$ Problem

The exact noisy-Tsirelson theorem solves one canonical path through Bell-correlator space. The full global problem remains:

$$C^*(\mu) = \sup \{|S(G)| : G \text{ is normalized PSD and } \mu\text{-submultiplicative on the Bell sector}\}.$$

A new Lean formalization decomposes this global problem into foundations, symmetry, bounds, lower bounds, and open conjectures. The project structure reported by the theorem generator is: `Defs.lean`, `Compactness.lean`, `Symmetry.lean`, `Bounds.lean`, `LowerBound.lean`, and `Summary.lean`.

6.1 Machine-checked global results

Result	Status	Meaning
<code>feasible_antitone</code>	machine-checked	If $\mu_1 \leq \mu_2$, then F_{μ_2} is contained in F_{μ_1} ; larger μ gives tighter constraints.
<code>CStar_antitone</code>	machine-checked	$C^*(\mu)$ is monotone decreasing in μ .

Extremizer existence	machine-checked	The optimization is well-posed on a compact feasible set.
negateS_feasible	machine-checked	A sign-flip symmetry preserves feasibility, so $ S $ can be reduced to maximizing S .
tsirelson_bound	machine-checked	PSD alone implies $S \leq 2\sqrt{2}$.
near_tsirelson_small_y	machine-checked	If $S \geq 2\sqrt{2} - \epsilon$, then $y^2 \leq 3\epsilon$.
strict_obstruction	machine-checked	For any $\mu > 0$, S cannot equal $2\sqrt{2}$.
large-correlator gap	machine-checked	If all $ x_{ij} > 1/2$, then $S^2 \leq 8 - \mu^2/8$.
simpleLB_feasible	machine-checked	$S=2$ is achievable for every μ in $[0,1]$.
Full $C^*(\mu)$	open	The tight global quantitative curve is not yet known.

These results establish a rigorous envelope: for $\mu > 0$, the supremum lies strictly below $2\sqrt{2}$, while the classical value 2 remains feasible. The monotonicity theorem makes the tradeoff mathematically precise even before the exact curve is known.

6.2 Conditional quantitative gap

The theorem engine proved a quantitative gap in the large-correlator regime. If all four Bell correlators satisfy $|x_{ij}| > 1/2$, then μ -submultiplicativity forces same-party overlaps to be nontrivial:

$$y^2 \geq \mu^2/16, \quad z^2 \geq \mu^2/16.$$

Together with the strengthened Tsirelson inequality $S^2 \leq 8 - 2y^2$, this gives

$$S^2 \leq 8 - \mu^2/8.$$

This is not yet a global quantitative gap because it assumes the Bell correlators are all large. It is nevertheless important: it proves that once a candidate extremizer is in the near-extremal large-correlator regime, nonzero μ produces an explicit gap below Tsirelson.

6.3 Failure of the naive symmetric ansatz

A natural first guess is the symmetric pattern $x_{00}=x_{01}=x_{10}=-x_{11}$ with $y=z$ chosen to saturate the μ constraints. Earlier numerical discussion treated the resulting expression $4/\sqrt{2+\mu}$ as a plausible ansatz. The new formalization shows that this construction is not a valid PSD Gram matrix for $\mu > 0$.

Specifically, for the commonly cited symmetric candidate with

$$x_{00}=x_{01}=x_{10}=-x_{11}=1/\sqrt{2+\mu}, \quad y=z=\mu/(2+\mu),$$

the Schur complement of the (A_0, A_1) block has determinant

$$-\mu^2/(2+\mu)^2 < 0 \quad \text{for } \mu > 0.$$

Thus the naive symmetric ansatz is infeasible. This explains why numerical searches favored asymmetric or higher-rank configurations. Finding tight lower bounds for $C^*(\mu)$ requires more sophisticated constructions.

7. Interpretation: What the Formal Status Means

The updated mathematical picture is sharper than the original. Geometry does not eliminate Bell nonlocality: the formal lower bound $S=2$ remains feasible for every μ , and weak or moderate nonlocality can coexist with positive consistency. But maximal Bell nonlocality cannot coexist with any nonzero uniform μ on the Bell sector.

The exact noisy-Tsirelson theorem solves the canonical visibility path into Tsirelson saturation. The global $C^*(\mu)$ theorem package proves monotonicity, compactness, symmetry reduction, strict obstruction, and a conditional quantitative gap, while also showing that the naive symmetric lower-bound construction is not PSD feasible.

The right conclusion is therefore not that the full tradeoff is solved. The conclusion is that the tradeoff is real, formally structured, and richer than the symmetric ansatz suggests.

8. Exponential-Decay Bridge: When μ Arises Physically

The μ -submultiplicativity condition is not only an abstract algebraic constraint. It arises naturally when correlations are approximately exponential functions of metric distance.

Theorem 8.1 (Exponential decay implies μ -submultiplicativity). Let (X, d) be a metric space. Suppose that for some $\xi > 0$ and $\epsilon \geq 0$,

$$\exp(-\epsilon) \exp(-d(i, j)/\xi) \leq |G(i, j)| \leq \exp(\epsilon) \exp(-d(i, j)/\xi)$$

for all i, j . Then

$$|G(i, k)| \geq \exp(-3\epsilon) |G(i, j)| |G(j, k)|.$$

Thus G is μ -submultiplicative with $\mu = \exp(-3\epsilon)$.

8.1 Proof

By the lower approximation bound, $|G(i, k)| \geq \exp(-\epsilon) \exp(-d(i, k)/\xi)$. The triangle inequality gives $d(i, k) \leq d(i, j) + d(j, k)$, hence $\exp(-d(i, k)/\xi) \geq \exp(-(d(i, j) + d(j, k))/\xi)$. The upper approximation bounds imply $\exp(-d(i, j)/\xi) \geq \exp(-\epsilon) |G(i, j)|$ and $\exp(-d(j, k)/\xi) \geq \exp(-\epsilon) |G(j, k)|$. Substituting gives the claimed $\exp(-3\epsilon)$ factor.

Corollary 8.2. If $|G(i, j)| = \exp(-d(i, j)/\xi)$ exactly for a genuine metric d , then G is 1-submultiplicative.

8.2 Physical interpretation

The bridge theorem identifies a broad physical regime where positive μ should be expected: whenever correlations decay approximately exponentially with metric distance. This includes many finite-correlation-length, transfer-matrix, thermal, diffusion-like, and gapped local systems. It gives the Bell-sector obstruction a physical reading: correlations compatible with exponential metric geometry cannot simultaneously realize maximal CHSH nonlocality on the same sector.

9. Open Problems and Limitations

- Determine the full global tradeoff curve $C^*(\mu)$.
- Prove a global quantitative gap $S \leq 2\sqrt{2} - c\mu^\alpha$, or show the optimal rate is different.
- Unify the large-correlator quantitative regime with the complementary small-correlator regime.
- Construct tight asymmetric or higher-rank lower-bound families for $C^*(\mu)$.
- Extend $\mu_{\max}(E)$ from correlator tables to full probability behaviors with marginals and no-signaling constraints.
- Apply $\mu_{\max}(E)$ to real loophole-free Bell-test data, including uncertainty-aware $\mu_{\max}^\epsilon(E)$.
- Compute $\mu(G)$ in physical correlation kernels from spin chains, tensor networks, diffusion kernels, and open quantum systems.

- Explore signed or phase-sensitive variants of $d = -\log|G|$ that retain more information than absolute correlation strength.

The main limitation is deliberate: the formal theorems are statements about normalized PSD kernels and correlator-level Gram completions. They do not by themselves constitute a full operational theory, a derivation of quantum mechanics, or a theory of spacetime. The value of the framework is that it provides a controlled mathematical language in which geometry, causality, and nonlocality can be compared.

10. Next Theorem Target

The theorem generator identified the remaining difficulty for a global quantitative bound. The large-correlator theorem already gives $S^2 \leq 8 - \mu^2/8$ when all $|x_{ij}| > 1/2$. The missing step is to prove that any near-Tsirelson extremizer must enter this large-correlator regime, or else to develop a separate gap theorem for the complementary regime.

A natural next target is: if $S \geq 2\sqrt{2} - \epsilon$ for ϵ sufficiently small, then each of x_{00} , x_{01} , x_{10} , and $-x_{11}$ is greater than $1/2$. Combining such a lemma with the large-correlator bound would produce a local quantitative gap near Tsirelson.

11. Conclusion

The revised framework now contains a stronger and more honest mathematical structure. It proves a general strict obstruction to Tsirelson saturation under nonzero geometric consistency, defines an observable completion invariant for Bell data, solves the noisy-Tsirelson completion path exactly, grounds μ -submultiplicativity in exponential metric decay, and formalizes a substantial part of the global $C^*(\mu)$ problem.

The most important correction is the failure of the naive symmetric ansatz. The optimal geometry-CHSH tradeoff is not captured by a simple symmetric shrinkage of the Tsirelson pattern. This makes the remaining problem more interesting: the true curve appears to depend on asymmetric or higher-rank Gram geometry. The resulting picture is a spectrum, not a binary: geometry and Bell nonlocality can coexist, but maximal Bell nonlocality forces a collapse of uniform relational geometry.

References

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Lean artifact: `RequestProject/NoisyTsirelsonCompletion.lean`, machine-checked exact noisy-Tsirelson completion theorem.

Lean artifact: RequestProject/Defs.lean, Compactness.lean, Symmetry.lean, Bounds.lean, LowerBound.lean, Summary.lean, machine-checked $C^*(\mu)$ formal status and symmetric-ansatz failure.

Bridge theorem added in this draft: approximate exponential metric decay implies μ -submultiplicativity with $\mu = \exp(-3 \epsilon)$.